EELU
الجامعة المصرية للتعلم الإلكتروني الأهلية
THE EGYPTIAN E-LEARNING UNIVERSITY

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Information Technology

IT438

Communication Technology

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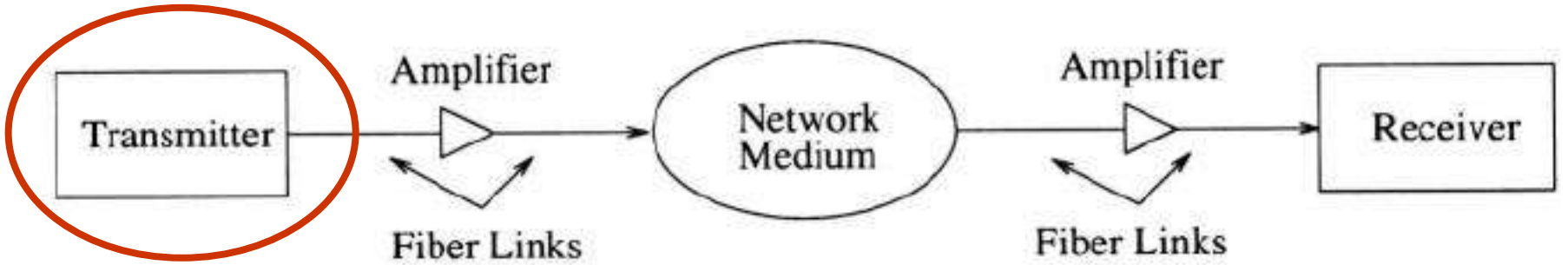
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Acknowledgement: *This presentation contains some figures and text from Data Communications and Networks book by W. Stallings*

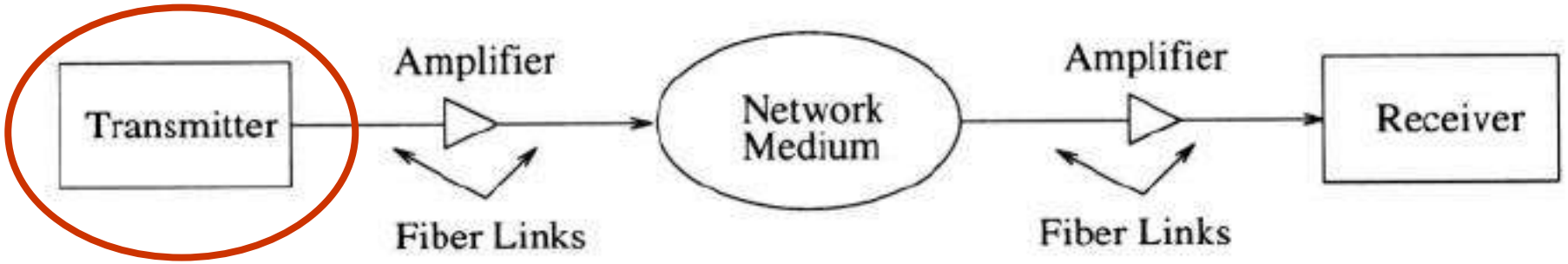
Fiber Optics Communication Technology

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Transmitters



The Laser



- In order to understand optical transmitters we need to understand the basic concepts of lasers.
- The word laser is an acronym for *Light Amplification by **Stimulated Emission** of Radiation.*
- Stimulated emission: allows a laser to produce intense high-powered beams of coherent light (light that contains one or more distinct frequencies).

The Laser *(cont.)*

- To understand the concept of stimulated emission, we must get some basic idea on the energy levels of atoms.
- In each atom, there are a number of discrete levels of energy that an electron can have (states).
- Stable Atoms: are in the ground state. In such a state, atoms have electrons that are in the lowest possible energy levels.
- To change the level of an electron in the ground state, the atom must *absorb energy*.

The Laser *(cont.)*

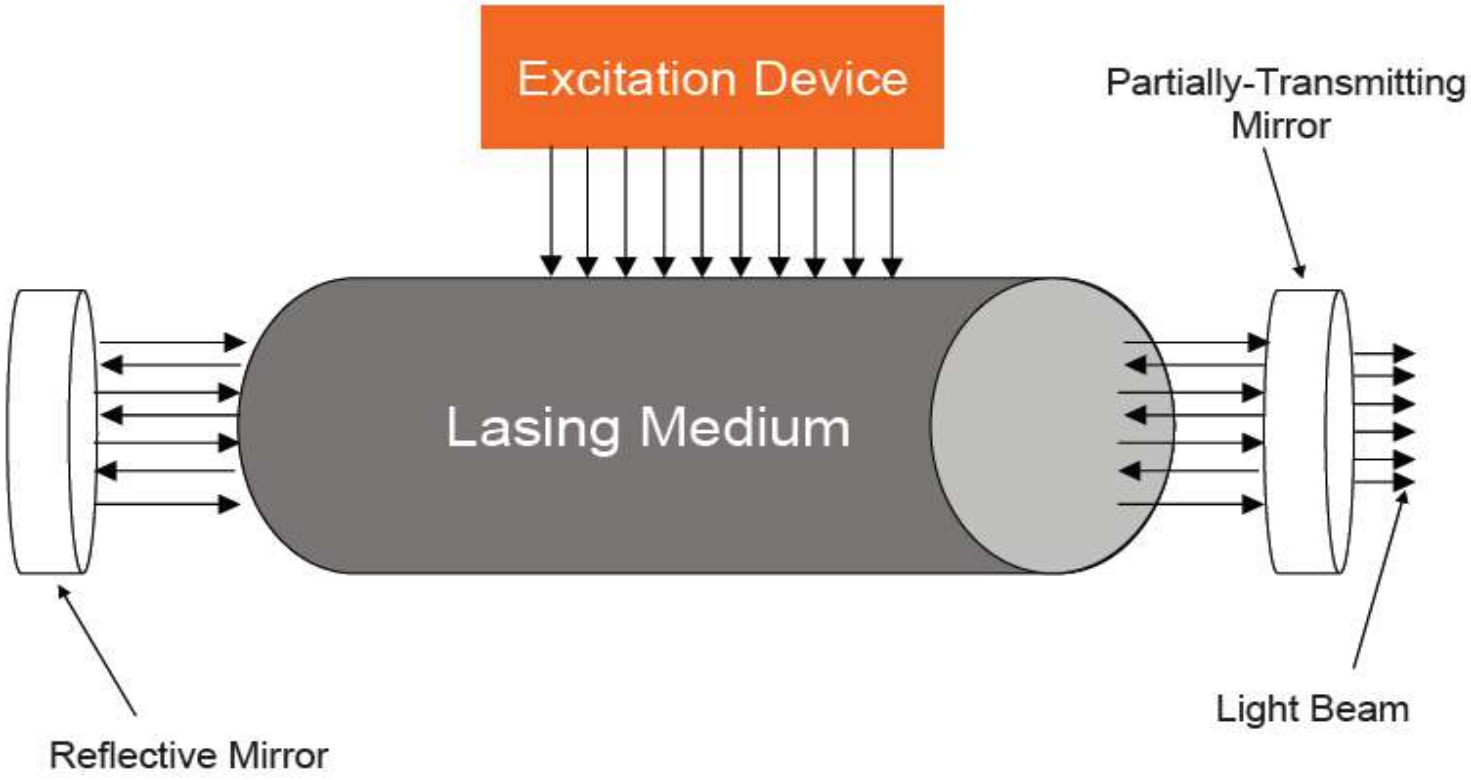
- When an atom absorbs energy, it becomes excited, and moves to a higher energy level.
- At this point, the electron is unstable, and usually moves quickly back to the ground state by releasing a *photon* (a particle of light).
- Quasi-stable states: the substance can stay in the excited state for longer periods of time.
- Applying enough energy to a substance in quasi-stable states for a long enough period of time causes *population inversion.*

The Laser *(cont.)*

- *Population inversion:* a state which means that there are more electrons in the excited state than in the ground state.

- Population inversion allows the substance to emit more light than it absorbs.

The Laser *(cont.)*



- Lasing medium is made of a quasi-stable substance.

The Laser *(cont.)*

- Excitation device excites electrons in the lasing medium.
- When an electron drops back to the ground state it emits a photon of light.
- The photon will reflect off the mirrors at each end of the cavity, and will pass through the medium again.
- Stimulated emission occurs when a photon passes very closely to an excited electron.

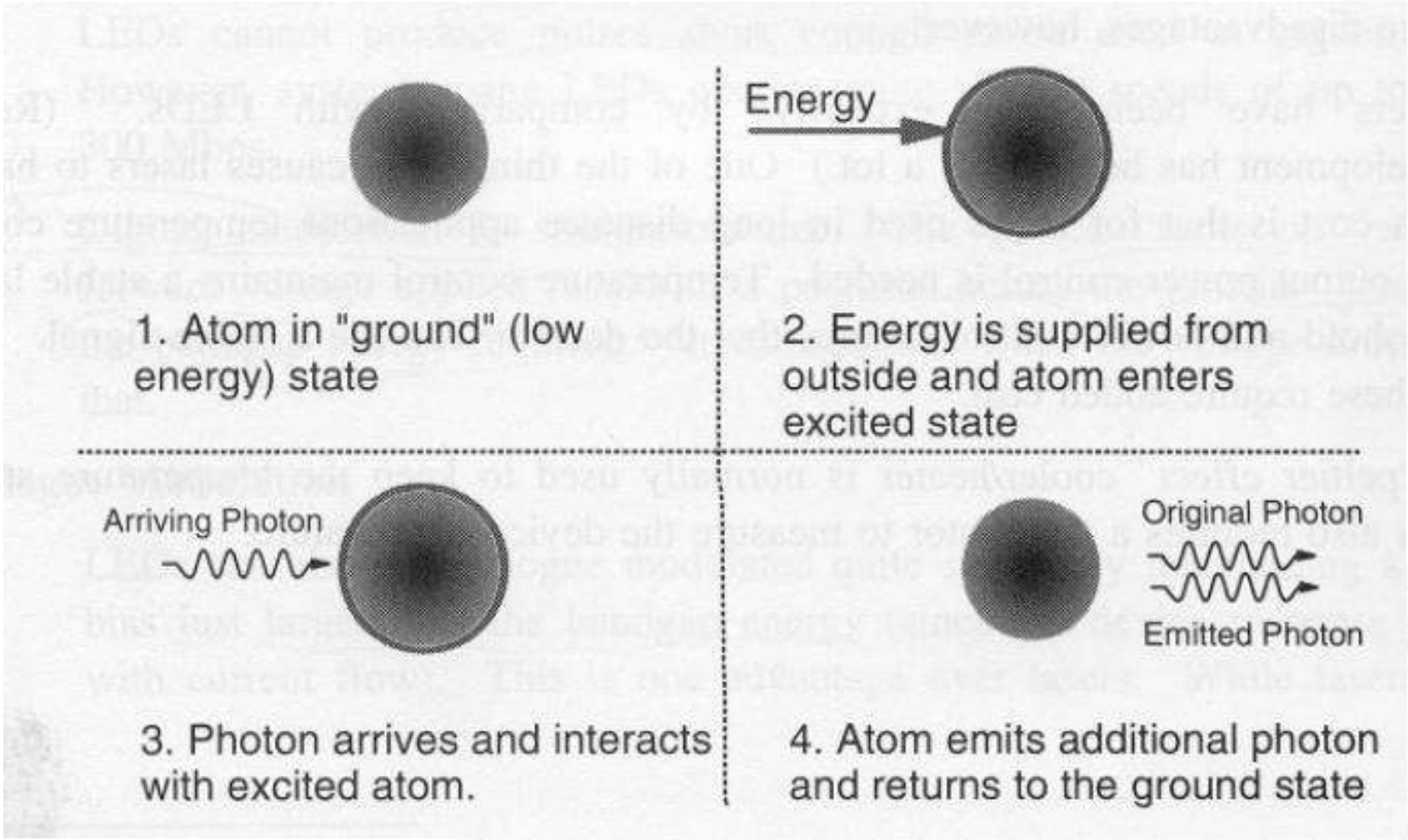
The Laser *(cont.)*

- The photon may cause the electron to release its energy and return to the ground state: a coherent photon is released.

- The process continues and the light at the selected frequency builds in intensity.

- The length of the cavity controls the frequency of the emitted light.

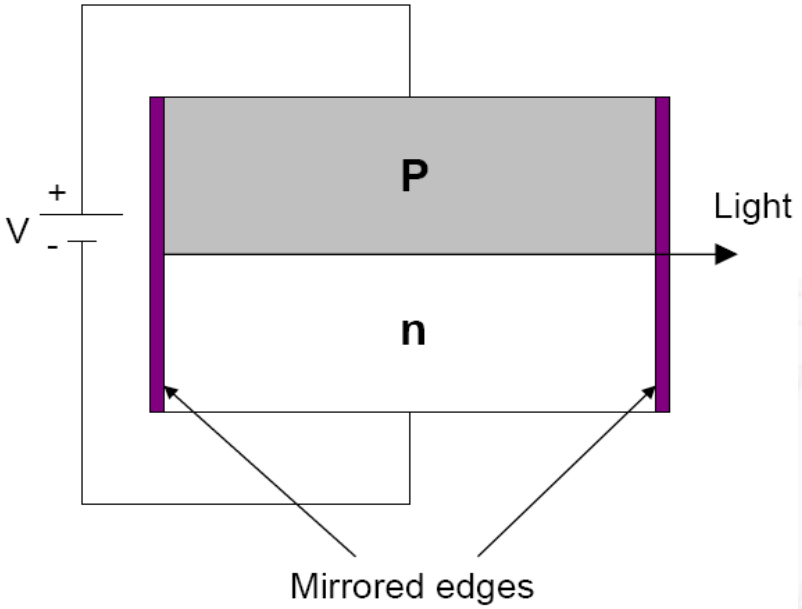
The Laser *(cont.)*



Semiconductor Diode Lasers

- Very simple and useful for optical communication.

- Can be simply implemented using a p-n junction with mirrored edges perpendicular to the junction.



Main Characteristics of Lasers

- Some of the physical characteristics of lasers which may affect system performance are:

  - laser linewidth,
  - Frequency instability, and
  - the number of longitudinal modes.

- Laser linewidth: is the spectral width of the light generated be the laser.

Main Characteristics of Lasers *(cont.)*

- Frequency instabilities: are variations in the laser frequency. Important types:
  - mode hopping
  - mode shifts

- The number of longitudinal modes: is the number of wavelengths that a laser can amplify.
  - Unwanted longitudinal modes may result in significant dispersion.
  - It is desirable to implement lasers which produce only a single longitudinal model.

Tunable Lasers

- Transmitters used in Wavelength Division Multiplexing (WDM) networks often require the capability to tune to different wavelengths.

- <u>Tunability characteristics:</u>
  - Tuning range: Wide and continuous range of over 100 nm
  - Tuning time: Rapid (< ms ranges)
  - Tuning capability: continuously vs. discretely tunable
  - Long lifetime and stable over lifetime
  - Easily controllable and manufacturable

Types of Tunable Lasers

- Tunable lasers can be implemented using different methods:
  - Mechanically-Tuned lasers
    - Physically adjust the distance between the two mirrors on either end of the cavity
    - Wide tuning range
    - Slow tuning time (order of ms)
  - Acousto-optic and Electro-optic -Tuned lasers
    - Use either sound waves or electrical current
    - Moderate tuning range
    - Moderate tuning time (order of tens of ns)
  - Injection-Current-Tuned lasers
    - Main drawback: mode hopping

Types of Tunable Lasers *(cont.)*

Tunable Transmitter	*Approx. Tuning Range (nm)*	*Tuning Time*
Mechanical (external cavity)	500	1-10 ms
Acoustooptic	83	10 μs
Electrooptic	7	1-10 ns (estimated)
Injection-Current (DFB and DBR)	10	1-10 ns

Laser Arrays

- Instead of using tunable lasers, one can implement a laser array, which contains a set of *fixed-tuned* lasers.
- A laser array consists of a number of lasers which are integrated into a single component, with each laser operating at a different wavelength.
- The advantage of using a laser array is that, if each of the wavelengths in the array is modulated independently, then multiple transmissions may take place simultaneously.
- A possible drawback: the number of wavelengths is fixed.



THANK YOU

QUESTIONS?





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